PHYC10009 Physics Foundations

Therapeutic uses of radiation, Nuclear Fission, Nuclear Fusion
Week 12 Lecture 1

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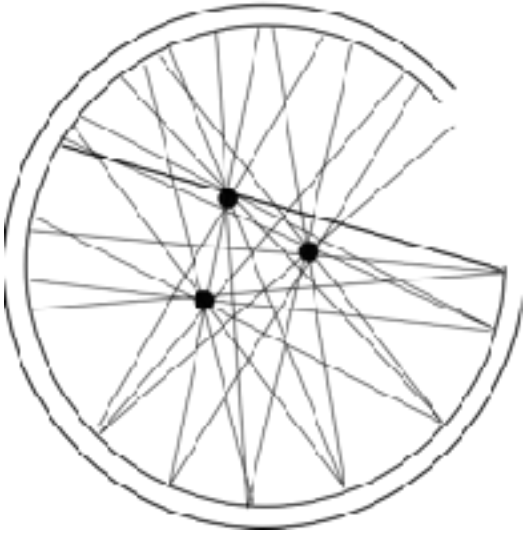
Summary

Contrast X-rays

The probability of X-ray absorption increase as Z^2 where Z is the atomic number of the material

Computerised X-ray Tomography (CT-scans)

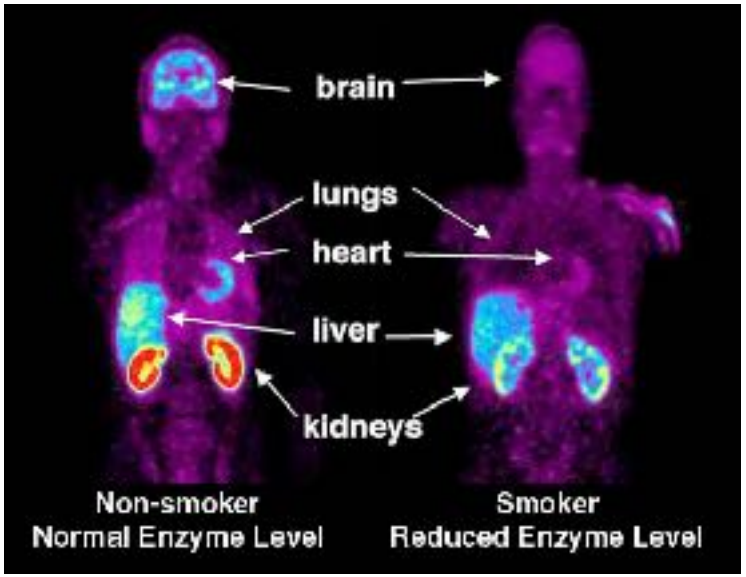


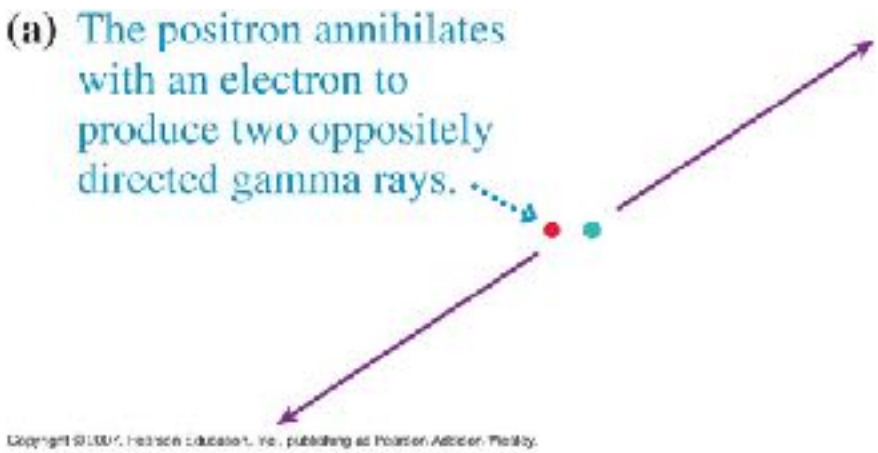
The intersection of response lines show the locations of Increased density

Contrast CT-Scans

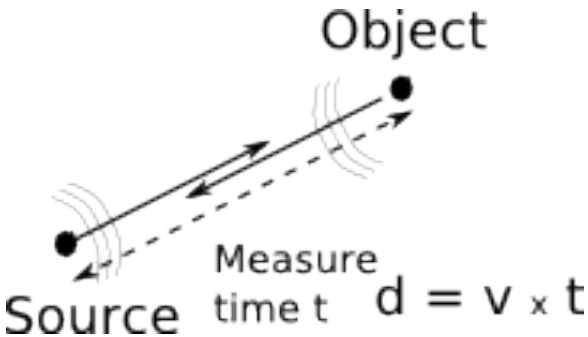
Summary continued...

Positron Emission Tomography (PET-scans)

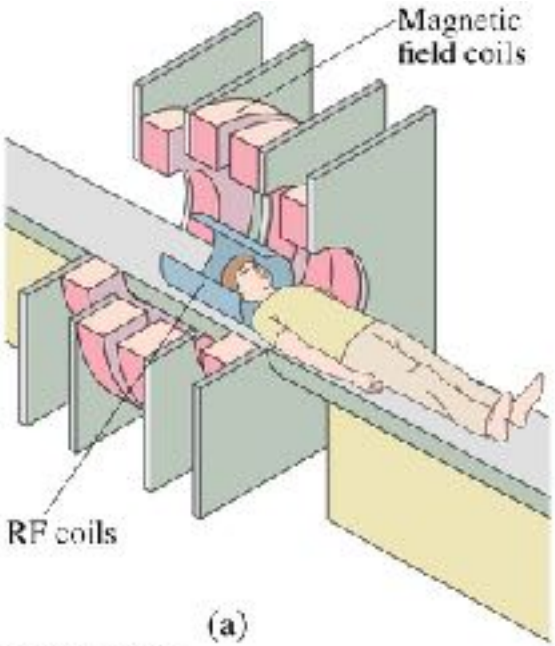




Ultrasound Imaging



Nuclear Magnetic Resonance Imaging (MRI) G31.9



(a)

(b)

Patient is placed in a strong Magnetic Field (~1 Tesla)
The area is bathed in radio waves near the energy difference between protons aligned and opposite the magnetic field

The magnetic field is adjusted to make a small region precisely match this energy difference.

Protons in this region flip to point opposite the magnetic field

They then fall-back to become aligned, emitting radio-waves as they do

These radio waves are picked up by a receiver. The strength of the signal received gives the proton density at that point in space.

Summary of non-invasive imaging techniques

Here is a summary of the medical imaging techniques we have discussed.

Technique	Comments	Resolution
Conventional X-ray	Cheap and simple. Very useful for bones. Small dose	0.5mm
CT scan, X-ray	3-D view of bone structures. Larger dose.	0.5mm
Contrast imaging, X-ray	Useful for imaging digestive tract. Small Dose	0.5 mm
Nuclear medicine, SPECT	Useful for imaging areas of high metabolic activity, especially bones. High Dose	1 cm
PET (positron emission)	Useful for monitoring metabolic activity. Particularly useful for Heart attack and stroke assessments High Dose.	2-5 mm
MRI	Useful for soft-tissue measurements. Measurements of chemical activities. No Dose.	0.5 mm
Ultrasound	Useful for soft-tissue/hard tissue interfaces. Particularly fetal development. No Dose.	0.3 – 2 mm

Todays lecture

- Radiation therapy
- Nuclear Fission
- Chain reactions
- Nuclear Fusion

Radiation Therapy - G31.6

Cells are most sensitive to radiation while they are reproducing so cancerous cells are preferentially killed by radiation.

Radiation therapy is a standard component of Cancer treatment.

- Maximise dose (and damage) to the target tumour and minimise radiation-induced morbidity to adjacent healthy tissues.
- Target a radiation beam to the tumour area through paths that spare nearby critical and radiosensitive anatomic structures;
- Select multiple fields that cross in the tumour area through different paths
- Partition the total dose in fractions (small amounts) over successive sessions.
- Healthy tissues recover better so each radiotherapy session provides accumulated cellular damage to the targeted tumour increases.
- Normal tissues are given the opportunity to repair.

Radiation therapy

Typically use gamma rays which are directed and collimated to target just the tumour site

Each treatment is designed to provide the maximum dose that can be sustained by the surrounding tissue.

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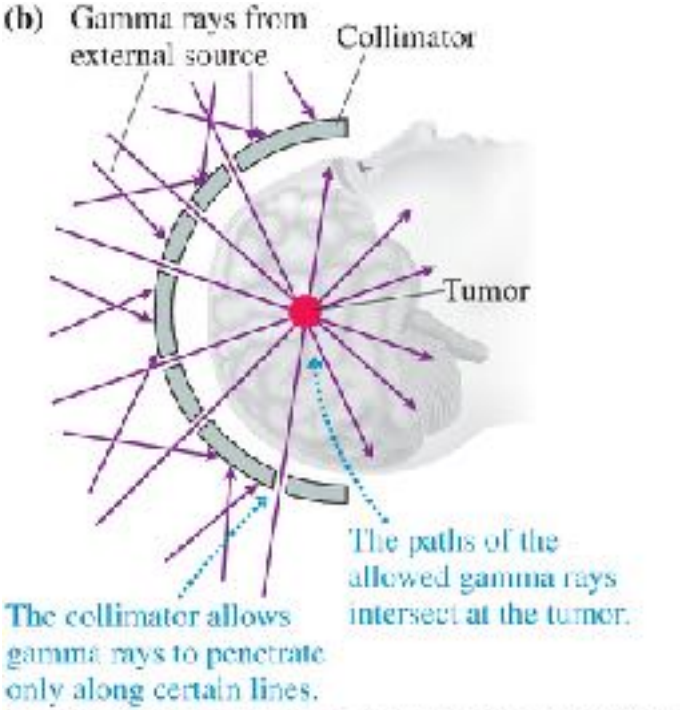

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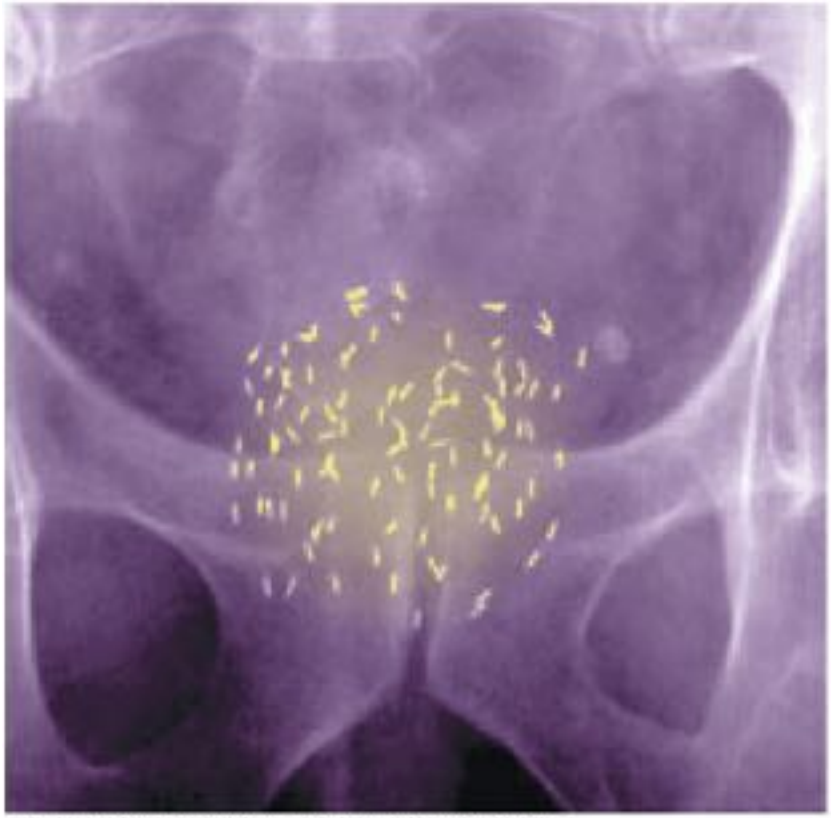
Radiation Therapy

Some tumours can be treated by the implantation of short-lived radioactive isotopes.

^{125}I is particularly effective since it emits photons of 25 KeV which have a very small range.

When placed in the region of interest these provide a large Dose to the tumour while limiting the dose to healthy tissue.



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Biological Half life

- Radio-active “tracers” enable the location concentration of biologically active molecules.
- The body breaks down and expels these as well
- This can the quantified with the concept of “biological half-life”
- Time at which half the molecule or isotope is removed
- For some long-lived radio-active isotopes, the biological half life defines the radiation dose to the body.
- Formula is the same as radio-active half-life:

$$N(t) = N_0 2^{\frac{-t}{t_{1/2}}}$$

$t =$ time

$N(t) =$ Number present at time t

$N_0 =$ Initial number present

$t_{1/2} =$ Biological Half-life

GAMSAT Question

Questions 103 and 104.
Radioactive materials have characteristic half-lives. In general, a material with a short half-life releases energy more quickly than a material with a long half-life.
If a radioactive material is introduced into the human body, the biological half-time is the time taken for the body to eliminate half the amount of radioactive material present at any time.
The radioactive half-life of a particular material is independent of its biological half-time.

103. A radioactive material has a half-life of 10 minutes.
How long will it take for 90% of the material to be transmuted?

A between 0 and 10 minutes
B between 30 and 40 minutes
C between 40 and 50 minutes
D between 90 and 100 minutes

104. In order to undergo a scan , a patient is injected with a radioactive isotope that has a half life of several days.

If 1% of the isotope remains in the patients body between 13 and 14 hours later, which of the following is the best estimate of the biological half-life of the isotope?

A 1 hours
B 2 hours
C 3 hours
D 4 hours

GAMSAT Question

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C between 40 and 50 minutes
D between 90 and 100 minutes

$$N(t) = N_0 2^{\frac{-t}{t_{1/2}}},\ 90\%\ \text{transmuted} \Rightarrow \frac{N(t)}{N_0} = 0.1 \quad \text{so:}$$

$$2^{\frac{-t}{t_{1/2}}} = 0.1 \Rightarrow \log_e 2^{\frac{-t}{t_{1/2}}} = \log_e 0.1 \Rightarrow -\frac{t}{t_{\frac{1}{2}}} = (\log_e 0.1)/(\log_e 2)$$

$$\Rightarrow t = -(10\ mins) \times \frac{-2.302}{0.693} = 33.2\ mins$$

(Ans B)

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A 1 hours
B 2 hours
C 3 hours
D 4 hours

$$N(t) = N_0 2^{\frac{-t}{t_{1/2}}},\ 1\%\ \text{remain} \Rightarrow \frac{N(t)}{N_0} = 0.01 \quad \text{so:}$$

$$2^{\frac{-t}{t_{1/2}}} = 0.01 \Rightarrow \log_e 2^{\frac{-t}{t_{1/2}}} = \log_e 0.01 \Rightarrow -\frac{t}{t_{\frac{1}{2}}} = (\log_e 0.01)/(\log_e 2)$$

$$\Rightarrow t_{1/2} = -t \times \frac{\log_e 2}{\log_e 0.01} = -\left(13.5\ hours\right) \times \frac{0.693}{-4.605} = 2.03\ hours$$

(Ans B)

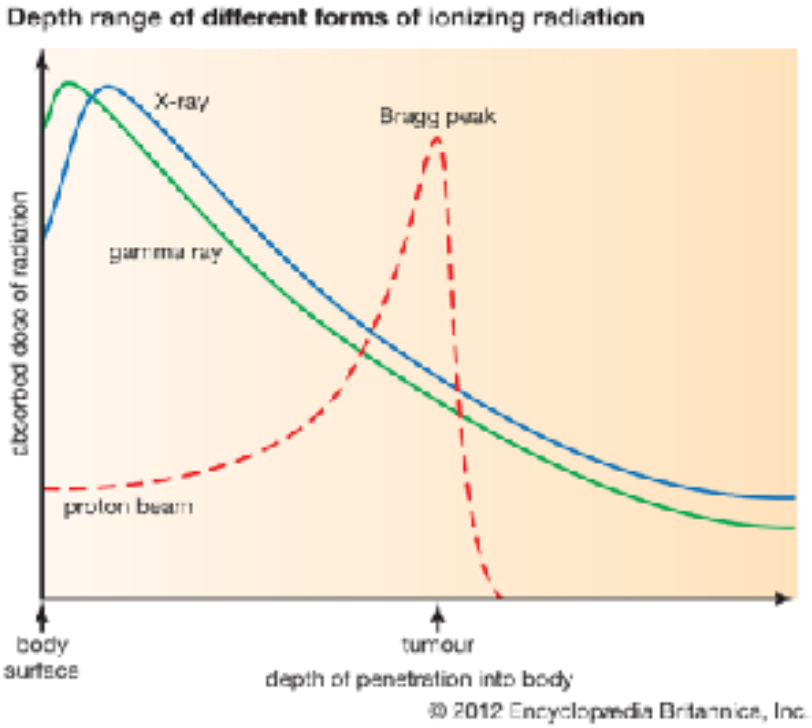
Charged particle radiation therapy

High energy charged particle beams like protons, ^{4}He and ^{12}C nuclei provide advantages for deep-seated tumours where the dose must be highly targeted.

As these particles travel through matter they slow down.

As they slow they are more effective at ionising matter.

Consequently the radiation dose is concentrated where the particles stop.



The radiation dose to tissues above and the side of the beam are smaller than gamma-ray based radiation therapy

But much more expensive and facilities are limited

Question 9, 2020 Exam

Fludeoxyglucose (FDG, ^{18}F) is a biologically active compound which is consumed in regions of the body with an active metabolism. After a heart attack, a patient is injected with -labelled FDG and scanned with a PET scanner to determine how much of the heart was damaged in the heart attack. ^{18}F is a radio-active isotope that decays via positron emission.

(a) Write the decay equation of showing the final state nucleus and the decay products of the decay of ^{18}F.

The initial activity of the compound injected into a 90 kg patient is 200×10^6 Bq. The half-life of is 110 minutes.

b) Calculate the number of nuclei injected into the patient.

c) Calculate the number of nuclei that decay in one hour.

d) The average energy deposited in the patient per decay is 0.4 MeV. Calculate how much energy is deposited in the patient in 1 hour.

e) The RBE of the positron emission is 1.5. Calculate the whole-body dose to the patient after all the nuclei have decayed.

Question 9, 2020 Exam, Solutions

(b) $R = \lambda N = \dfrac{0.693N}{t_{1/2}} \Rightarrow N = \dfrac{2 \times 10^8 \times t_{1/2}}{0.693}$

$$= \frac{2 \times 10^8 \times 110 \times 60}{0.693} = 1.9 \times 10^{12}$$ ^{18}F nuclei

c) $N(t = 60min) = \left(\dfrac{1}{2}\right)^{\frac{60}{110}} \times 1.9 \times 10^{12} = 1.3 \times 10^{12}$

N_{decays}= N_0 – N(t=60 min) = $1.9 \times 10^{12} - 1.3 \times 10^{12} = 6.0 \times 10^{11}$

d) E = $N_{decays} \times$E/decay = $6.0 \times 10^{11} \times 0.4 \times 1.6 \times 10^{-13} = 0.0384 J$

e) Dose = RBE $\times E_{deposited}$/mass

$= 1.5 \times 1.9 \times 10^{12} \times 0.4 \times 1.6 \times 10^{-13}/90 = 2\ mSv$

Nuclear Reactions G31.6

A nuclear reaction takes place when a nucleus is struck by another nucleus or particle.

If the original nucleus is transformed into another, this is called transmutation.

An example: $${}^{4}_{2}\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H},$$

Energy and momentum must be conserved in nuclear reactions.

Generic reaction: $$\mathrm{a} + \mathrm{X} \rightarrow \mathrm{Y} + \mathrm{b},$$

The reaction energy, or *Q*-value, is the sum of the initial masses less the sum of the final masses, multiplied by c^2:

$$Q = (M_{\mathrm{a}} + M_{\mathrm{X}} - M_{\mathrm{b}} - M_{\mathrm{Y}})c^2.$$

If *Q* is positive, the reaction is exothermic, and will release energy.

If *Q* is negative, the reaction will absorb energy.

GAMSAT Questions 28 - 30

Positron emission tomography (PET) is used in many major hospitals as a diagnostic tool. Positrons are similar to electrons but have a charge of +1 and so their symbol is . A radioactive isotope of fluorine, fluorine-18, is commonly used as the positron emitter. This isotope is produced in a cyclotron by bombarding oxygen-18 with protons.

28 Which one of the following nuclear equations represents the emission of positrons from fluorine-18?

A ${}^{18}_{9}F \rightarrow {}^{18}_{8}O + {}^{0}_{1}e$
B ${}^{18}_{9}F \rightarrow {}^{18}_{8}F + {}^{0}_{1}e$
C ${}^{18}_{9}F \rightarrow {}^{18}_{10}Ne + {}^{0}_{1}e$
D ${}^{18}_{9}F \rightarrow {}^{17}_{10}F + {}^{0}_{1}e$

29 Radiation doses are calculated by the energy deposited per kilogram of body tissue, for which the SI unit is the gray (Gy).

$1\ Gy = 1\ Jkg^{-1}$

The average energy from a single annihilation is approximately $10^{-13}\ J$ and there is an average of 10^{9} annihilation per second over the period of 20 minutes needed for a PET scan.

What is the best estimate of the whole-body dose received by a 60 kg person during a PET scan?

A 2.0 Gy
B 0.20 Gy
C 0.020 Gy
D 0.0020 Gy

GAMSAT Questions 28 - 30

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C $^{18}_{9}F \rightarrow ^{18}_{10}Ne + ^{0}_{1}e$
D $^{18}_{9}F \rightarrow ^{17}_{10}F + ^{0}_{1}e$

Positron emission is the process of converting a proton to a neutron. So the number of nucleons does not change but the isotopes' proton number decreases by one. So right answer is **A**

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A 2.0 Gy
B 0.20 Gy
C 0.020 Gy
D 0.0020 Gy

The radiation dose = $10^{-13}J \times 10^{9}Bq \times 20\ mins \times \frac{60\ s/min}{60\ kg} = 0.002\ Gy$

Ans **D**

GAMSAT Question 30

30 Which one of the following nuclear equations represents the production of radioactive fluorine in a cyclotron?

A ${}^{18}_{9}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{0}_{1}e$

B ${}^{18}_{9}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{0}_{-1}e$

C ${}^{18}_{9}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{1}_{1}H$

D ${}^{18}_{9}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{1}_{0}n$

GAMSAT Question 30

30 Which one of the following nuclear equations represents the production of radioactive fluorine in a cyclotron?

A ${}^{18}_{8}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{0}_{1}e$

B ${}^{18}_{8}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{0}_{-1}e$

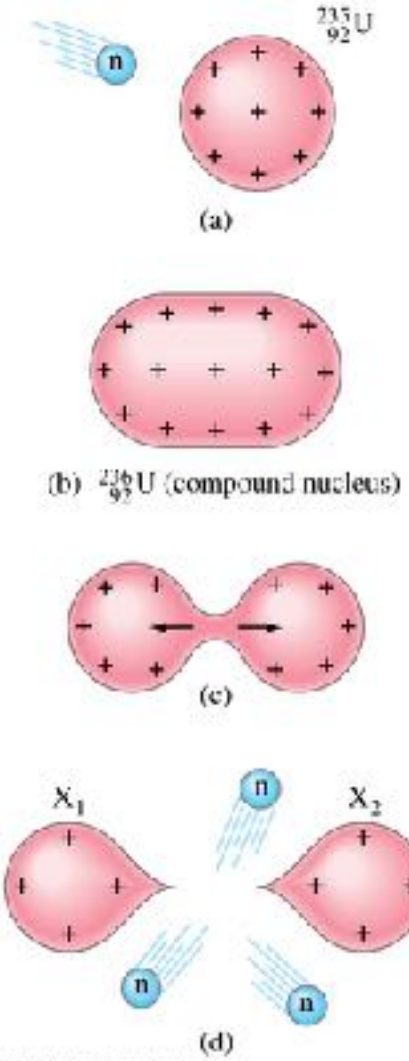
C ${}^{18}_{8}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{1}_{1}H$

D ${}^{18}_{8}O + {}^{1}_{1}H \rightarrow {}^{18}_{9}F + {}^{1}_{0}n$

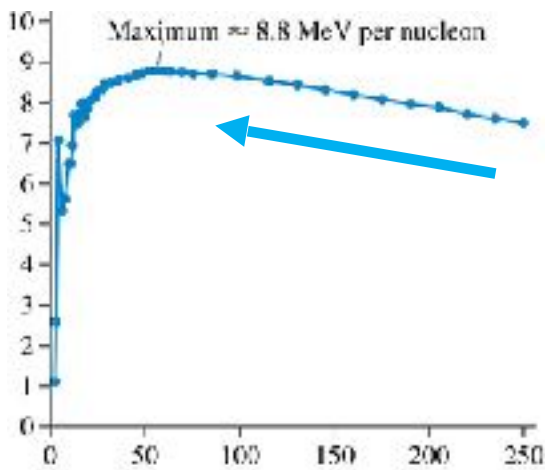
Need to balance proton and neutron numbers of both sides of the reaction while creating . So answer is **D**

Nuclear Fission G31.2

- Nuclear fission occurs when a heavy nucleus like ^{235}U absorbs a neutron
- After absorbing a neutron, a uranium-235 nucleus will split into two roughly equal parts.
- One way to visualise this is to view the nucleus as a kind of liquid drop.
- In the process it releases an enormous amount of energy ~200 MeV
- In comparison a strong chemical reaction like $H_2 + O \rightarrow H_2O$ releases 18 eV, **10 million times less.**
- Furthermore, after the fission 2-3 other neutrons are released. These can initiate further reactions.



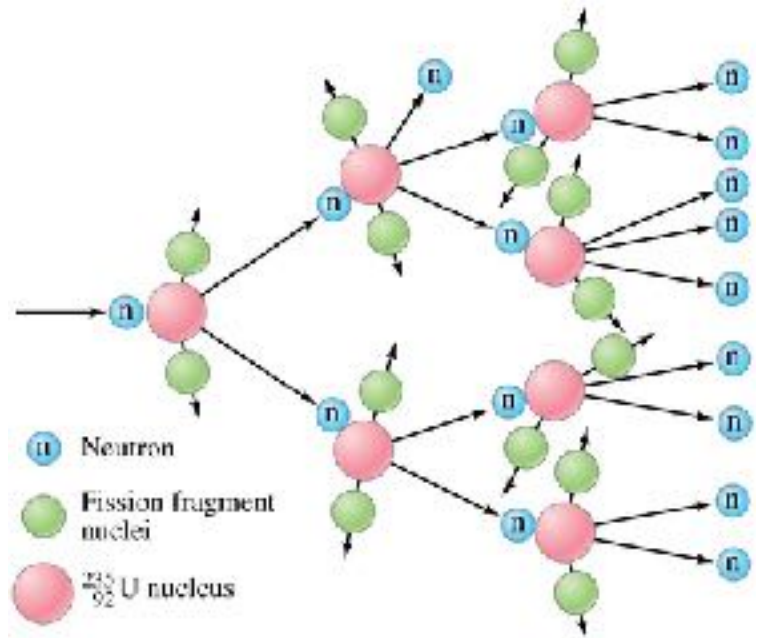
B.E./Nucleon



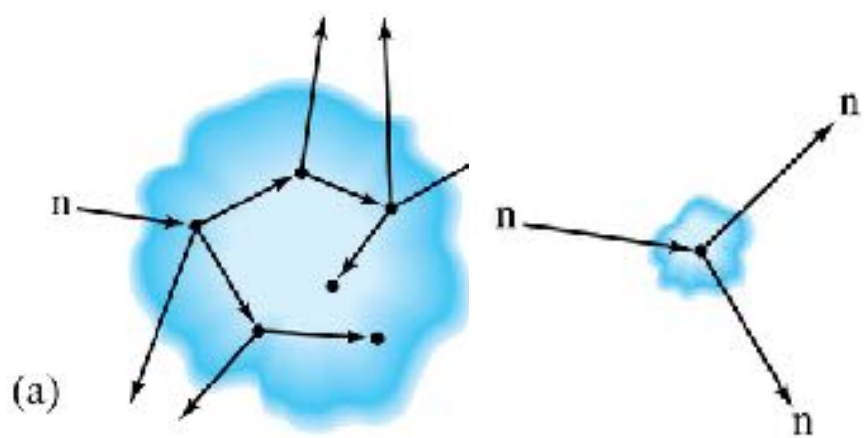
Fission increases binding energy by ~ 1 MeV/nucleon ~ 200 MeV

Nuclear Fission

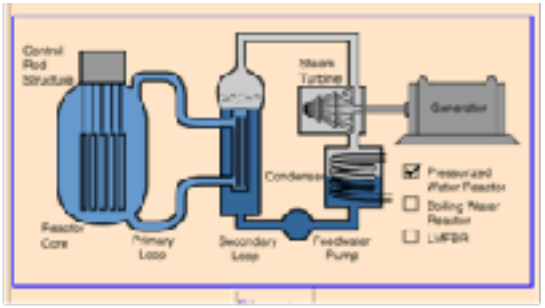
If a large quantity of fissionable nuclei are close together these neutrons can initiate a *chain-reaction* causing a macroscopic number of nuclei to fission and release energy.

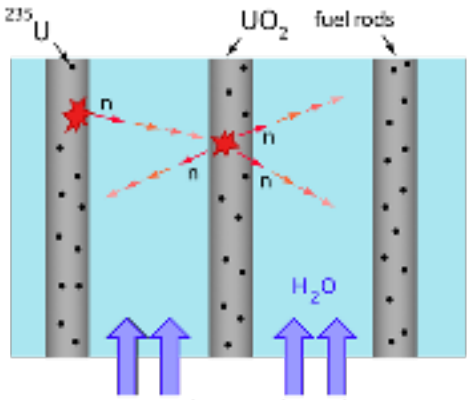


Neutrons that escape from the uranium do not contribute to fission. There is a critical mass below which a chain reaction will not occur because too many neutrons escape.

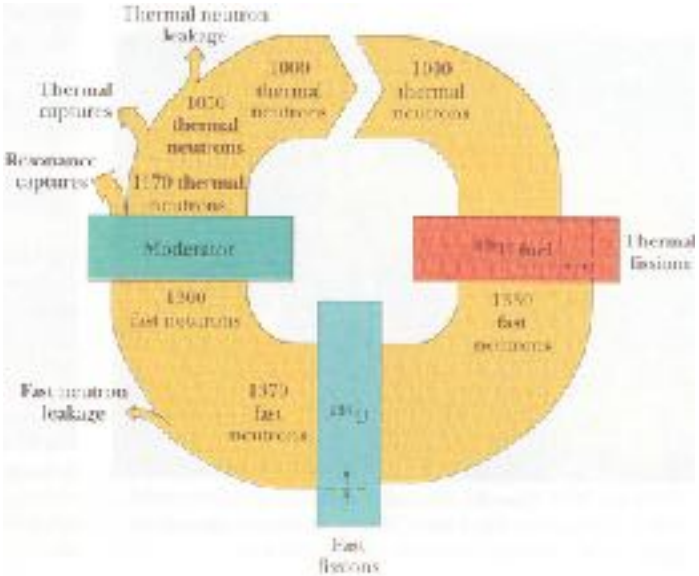


Light water nuclear reactors





Water as coolant and moderator flows between fuel rods.



The neutron multiplication factor is adjusted to be exactly 1.000000

Control of light water reactors:

- Delayed neutron emission
  0.7% neutrons emitted after beta decay (8 seconds)
- Negative temperature coefficient (k reduces with T)
- Negative “void” coefficient.
  Loss of coolant through bubble formation or other means, means no further moderation and a decrease in reactivity.

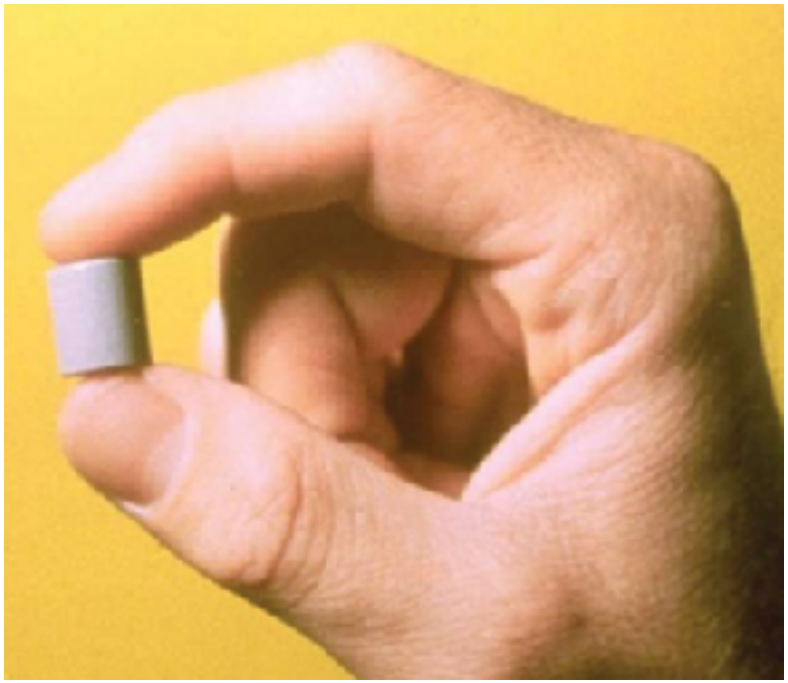
Nuclear Power

Energy required for 1 Australian Family/year.

12 tonnes of coal

Coal emits ~30 tonnes of CO_2 per family per year

10 grams of nuclear fuel



Nuclear power emits no CO_2 during operations

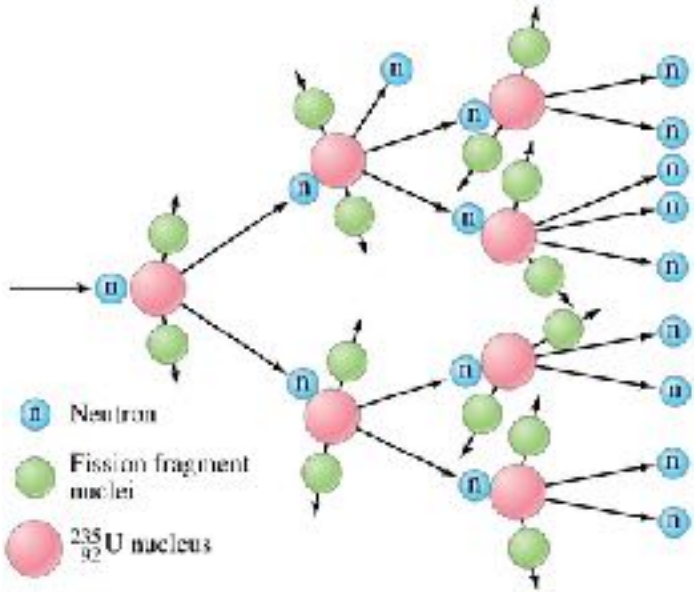
Safety of Nuclear Power

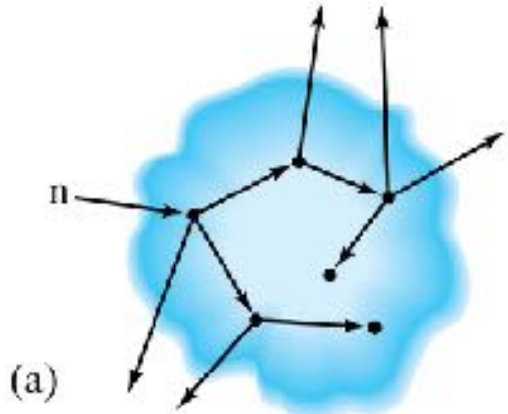
- Typical large Nuclear Power Plant contains 10 billion Giga-Becquerel's of activity.
- 1 Giga-Becquerel typically leads to an unwanted exposure.
- Nuclear Power Plants contain vast amounts of dangerous material.
- Effectively shielded by water and concrete.
- Safely handling this is a significant challenge and makes building Nuclear Power plants significantly more expensive than coal-fired power stations.
- Nuclear costs around $10 Billion/GW vs $5 Billion/GW for coal.
- On the other hand the nuclear fuel is significantly cheaper than coal in many parts of the world.

Uncontrolled chain reaction

If there is enough ^{235}U, ^{233}U or ^{241}Pu concentrated in a large mass, (*critical mass*), an uncontrolled chain reaction occurs.

A large fraction of the nuclei fission in microsecond resulting in an extremely powerful explosion – a nuclear weapon



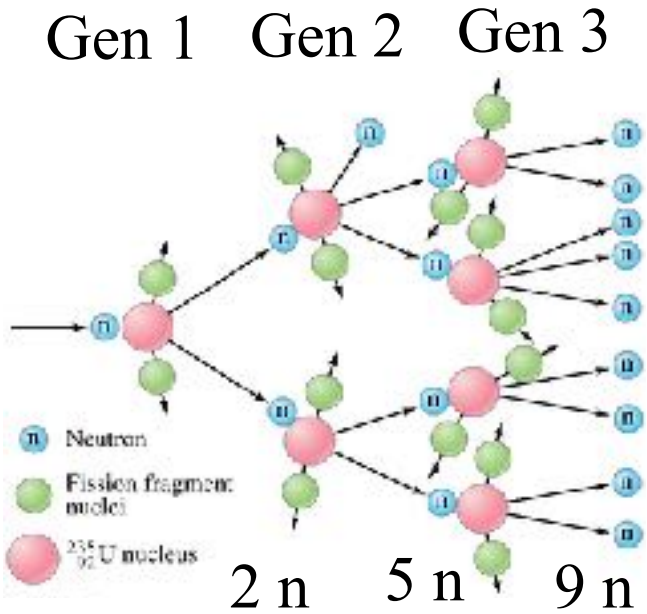


Mousetrap Demo

D Na-2 Nuclear Chain reaction

Exponential growth

A nuclear chain reaction is an example of exponential growth



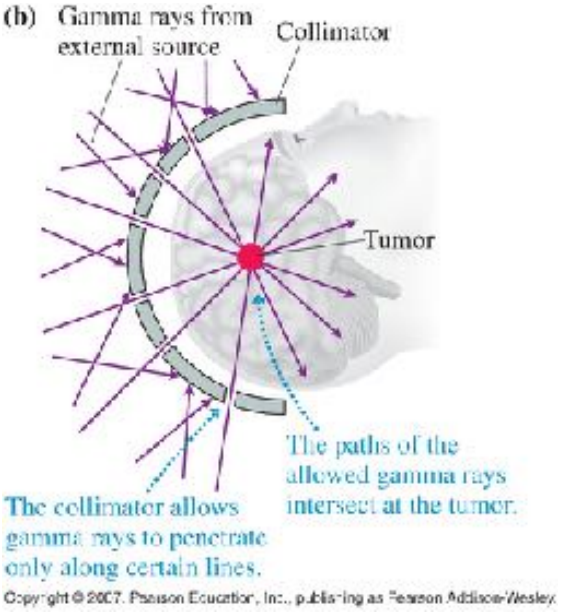
Reproduction time of each generation, R_t, and the number of neutrons in each generation , (the multiplication factor), k , describes how fast nuclei are consumed. Let N = Total number, N_0= Initial number, n = Generation number, then:

$$N(n) = N_0 k^n, \text{ Time} = n \times R_t$$

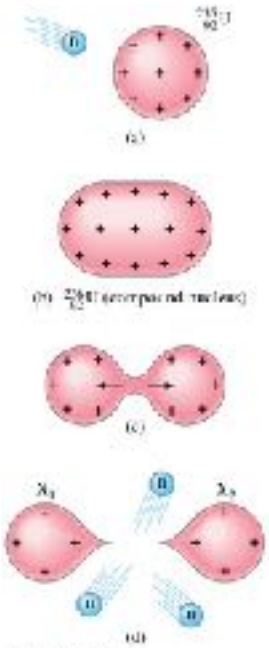
These are the same principles as the spread of Covid-19

Summary

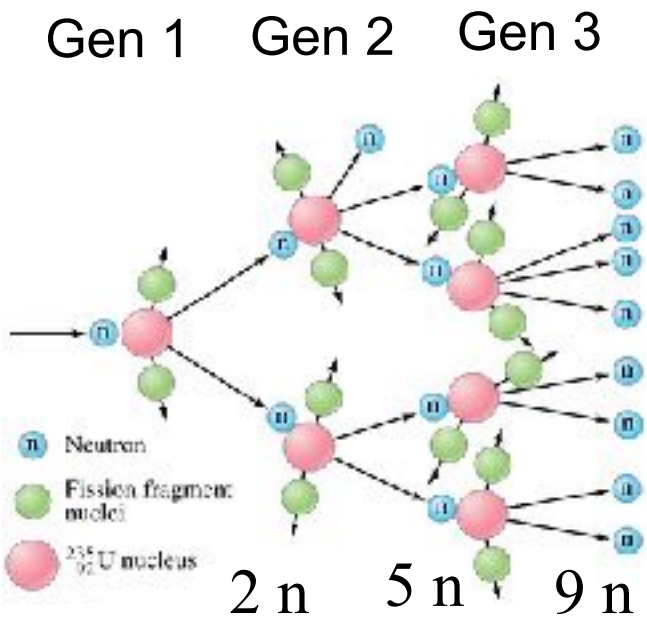
Radiation Therapy: apply the maximum dose to a tumour while minimising dose to healthy tissue.



Nuclear Reactions, Nuclear Fission



Applications of Nuclear fission:
Nuclear chain-reactions



$$N = N_0 k^n, \text{ Time} = n \times R_t$$

Exponential Growth
(Covid-19)